

NSLS-II. Photo: Brookhaven National Laboratory

WORKSHOP PROPOSAL · BROOKHAVEN NATIONAL LABORATORY · OPEN TO ALL STUDENTS

Turning a Chess Engine into Science AI

In one week, 20 students build a chess AI in plain Python, then aim it at a science problem modeled on Brookhaven's labs.

30	20	~760	0
class hours over 5 days, four 90-minute periods a day	students in 10 student pairs, coding and playing chess together	lines of Python each student pair types in and runs	student online accounts; everything runs offline after the Day 1 setup

Chess is where AI learned to think ahead. Science is where it is going next.

Every game-playing AI runs the same loop: describe the situation, list the possible actions, look ahead, score the results, and check the answer. An AI agent at a beamline runs that loop to choose its next measurement.

Students who know some coding and chess build Remus in plain Python. Student pairs type four engines and compare their code with stored reference results using verify_remus.py. Module runs check syntax and imports; passing the verifier covers its reference cases. Lucas Chess games use instructor engines and illustrate play; they do not test student files or establish a rating.

SAME LOOP, TWO WORLDS

	Represent	Generate	Look ahead	Evaluate	Verify
IN REMUS	An 8×8 board and the full game state	Every legal move, including en passant	Minimax and alpha-beta search	Score material and position	Perft counts match a trusted reference
IN SCIENCE AI	A sample, a model, or an instrument state	Candidate measurements or hypotheses	Predict outcomes before spending beam time	Score how well each hypothesis explains the data	Compare predictions with measured data

What students will be able to do AND HOW WE WILL KNOW

OUTCOME	EVIDENCE A REVIEWER CAN CHECK	WHEN
Model a system as data	Student Remus1 matches published perft counts at six reference positions	Day 1
Read and reproduce real code	Student copies match verifier cases; module runs check syntax and imports	Days 1-4
Reason about search	Student pairs explain reported depth, score, and nodes; no speed guarantee	Days 2-3
Measure a change	Instructor games illustrate play; they do not test student files or isolate an enhancement's effect	Days 2-4
Find a hidden peak	Compare readings on 100 generated samples; record parameter settings	Day 5
Name the crystals in a pattern	Compare right, wrong, and ambiguous counts on 200 generated powders; printed expectations require manual comparison	Day 5
Explain a result in plain words	Each student pair presents measured science results and parameter values	Day 5

The decision

1. **Approve a pilot** for 20 students, designed to grow to 30: one classroom for 5 days, 8:30 to 4:30, 10 Windows 11 laptops, and 20 USB drives.
2. **Arrange a scientist visit** on Day 5, and share one measured two-phase powder pattern from the PDF or XPD beamline for the phase hunt.
3. **Support an evaluation** with Brookhaven education staff: a concept check and an interest survey before and after the week.
4. **Judge it by evidence:** student code, games, science results, and explanations, not chess strength. No student accounts; code goes home on USB.